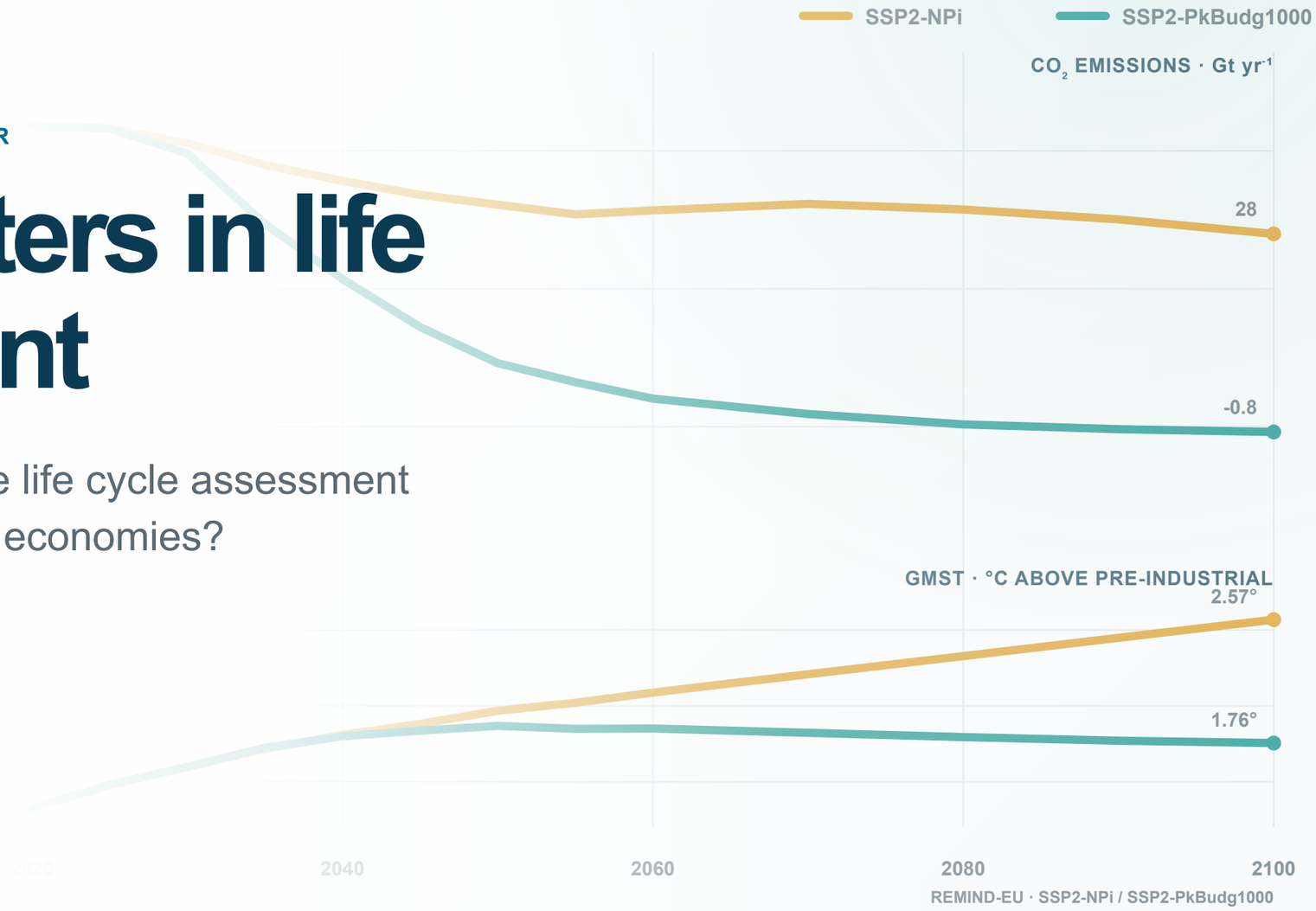
SEOUL NATIONAL UNIVERSITY · GRADUATE LCA SEMINAR

Why timing matters in life cycle assessment

How does accounting for time change the life cycle assessment of long-lived systems in rapidly changing economies?

Romain Sacchi

Paul Scherrer Institut · 30 October 2026



ABOUT ME

Romain Sacchi



Scientist

Technology Assessment Group

Laboratory for Energy Systems Analysis

Paul Scherrer Institut · Switzerland

PSI profile ↗



Path to PSI

2012

MSc

Environmental Sciences · Aalborg University

2018

Industrial PhD

Industrial Ecology · Aalborg University

2019

Joined PSI

Technology Assessment Group

Research interests

Industrial ecology

Future energy & mobility systems

Time & climate

Prospective + dynamic LCA

Open science

Transparent, usable modelling tools

Open-source projects

github.com/romainsacchi ↗

premise
Future LCA databases ↗

TRAILS
Time-explicit LCA ↗

carculator
Vehicle life-cycle models ↗

EDGES
Context-aware impact assessment ↗

Selected publications

Full record · ORCID ↗

Renewable & Sustainable Energy Reviews · 2022

premise: databases for prospective LCA

Read paper ↗

Nature Communications · 2023

How to make climate-neutral aviation fly

Read paper ↗

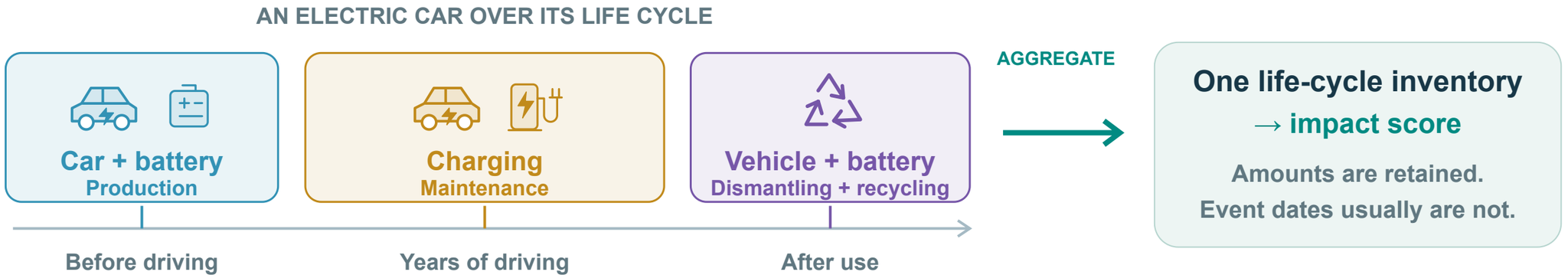
Renewable & Sustainable Energy Reviews · 2022

Electrification of passenger cars: when, where and how?

Read paper ↗

Conventional LCA usually collapses time

An electric car is manufactured, charged and maintained for years, then dismantled and recycled—all can enter one total without event dates.



01 · LONG-LIVED ACTIVITIES

Suppliers change during operation

The electricity used to charge the same car can change over its years of driving.

Long-lived plants and replacement equipment face the same issue.

Müller et al. (2025)

02 · CLIMATE EFFECTS

Climate depends on emission dates

CO_2 emitted now affects the climate before a later removal occurs. A single total hides that history.

Relevant to forest regrowth, weathering and delayed CO_2 removal.

Levasseur et al. (2010)

03 · DELAYED IMPACTS

Impacts can arise long after disposal

Landfill leachate can release pollutants centuries later, causing potential human-toxicity impacts.

A short horizon or a ‘no LT’ method can exclude these effects.

Turner et al. (2017) · ecoinvent LT guidance

Temporal inventory foundations: Beloin-Saint-Pierre, Heijungs & Blanc (2014)

Changing supply systems and carbon timing: four cases

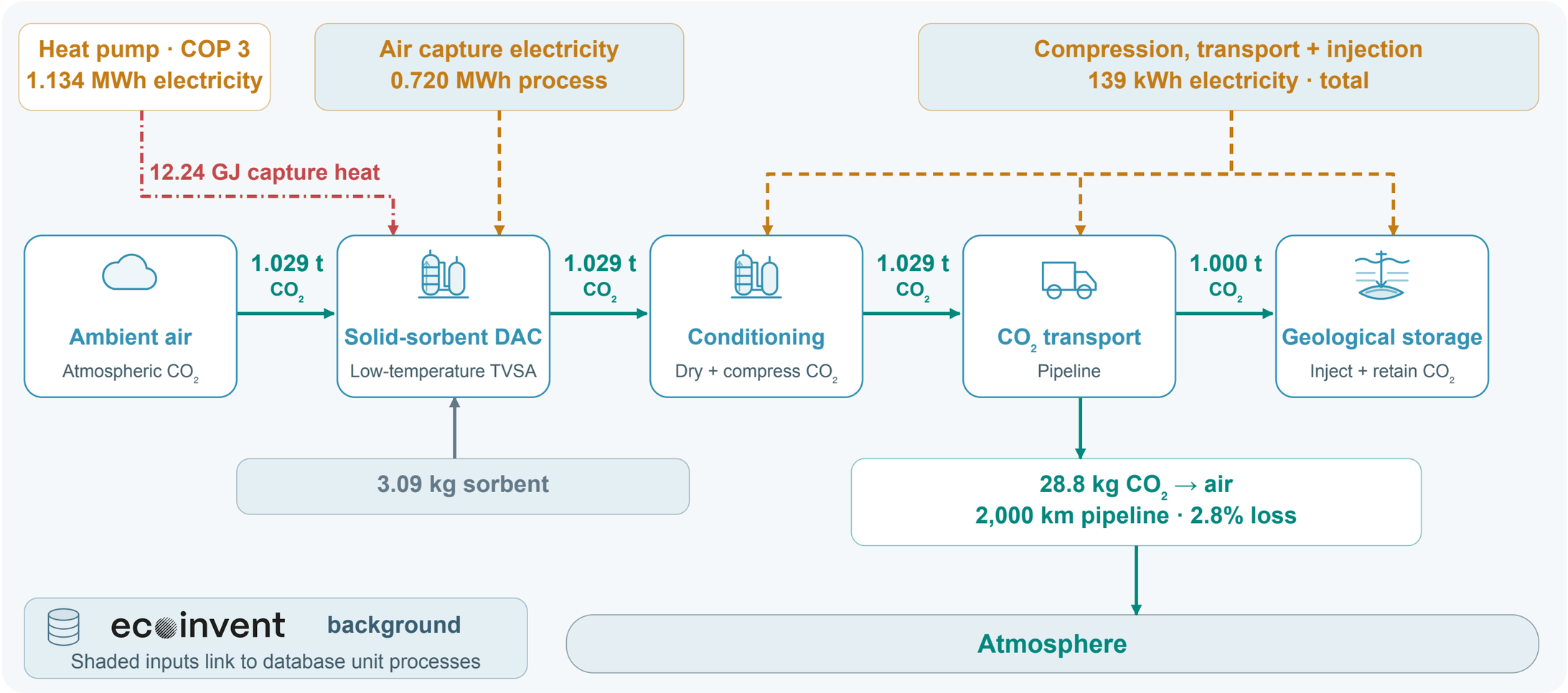
	CARBON SOURCE	INTERVENTION	DESTINATION
DACCS **1 t CO_2 physically stored** vs No DAC plant	 **Air**	 **Direct air capture**	 **Geological storage**
	Supply system: Electricity + heat	Timing: Build → capture during operation	
BECCS **1 t CO_2 physically stored** vs Standing forest + market energy	 **Biomass**	 **Energy + capture**	 **Geological storage**
	Supply system: Biomass + displaced energy	Timing: Harvest → regrowth over decades	
ERW **1.25 t rock deployed** vs No additional rock deployment	 **Air**	 **Alkalinity enables uptake**	 **Dissolved ocean carbon**
	Supply system: Grinding electricity + haulage	Timing: Deploy → dissolve → atmospheric uptake	
Cement + CCS **1 t CO_2 stored · CCS − BAU** vs Same clinker without capture	 **Limestone + fossil fuels** **Non-fossil fuels**	 **Kiln capture**	 **Geological storage**
	Supply system: Kiln fuels + capture utilities	Timing: Build → capture + avoided release	

Follow carbon, materials and energy to storage

DACCS

Low-temperature solid-sorbent DAC · TVSA

1 t CO_2 physically stored



Adapted inventory: Deutz & Bardow (2021) · Qiu et al. (2022) **CO_2 transport / storage:** Koornneef et al. (2008) · Terlouw et al. (2021)

Foreground systems modelled with Brightway · linked to ecoinvent background suppliers

Electricity — kWh or MWh

Heat — GJ of delivered heat

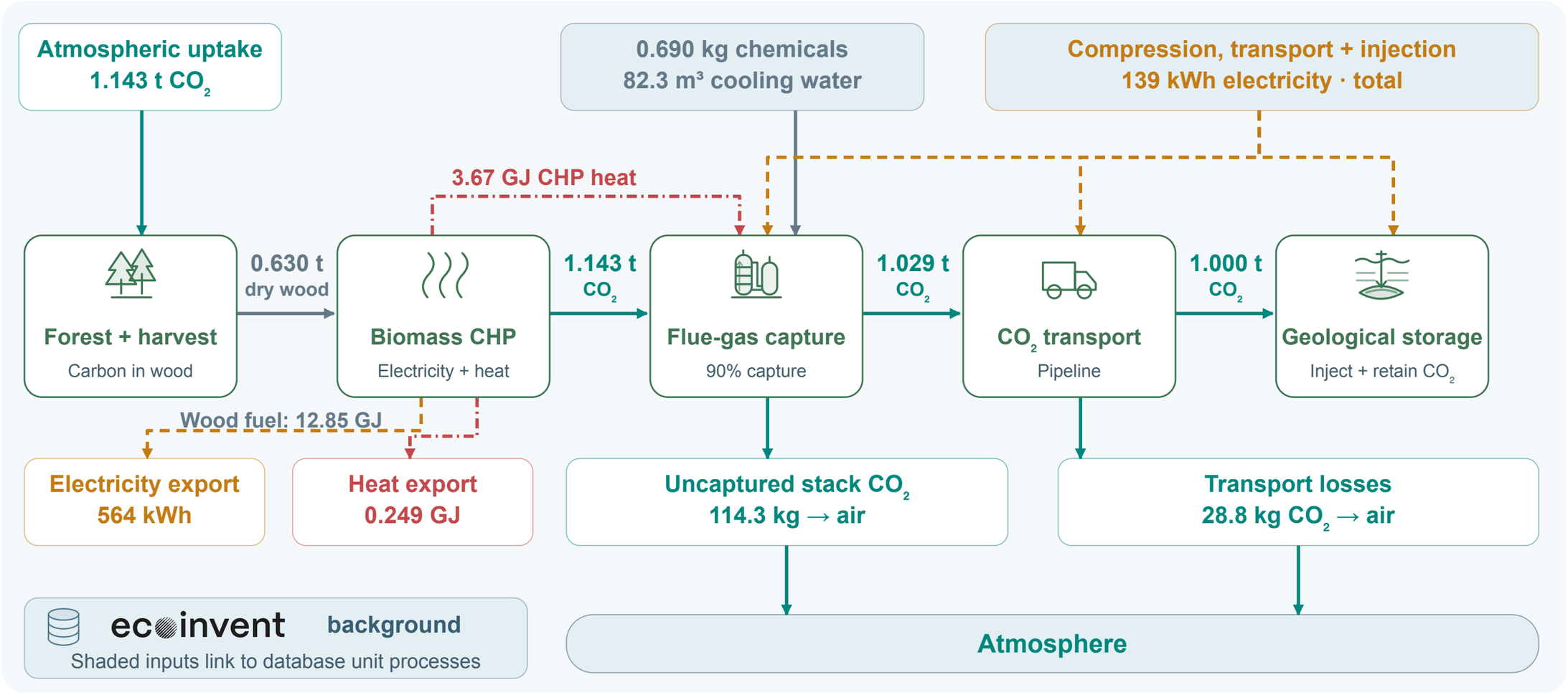
Materials + fuels — kg, t, m³ · fuel energy in MJ/GJ

CO_2 — kg or t · capture, storage, release

Follow carbon, materials and energy to storage

BECCS

1 t CO_2 physically stored



Adapted inventory: ecoinvent 3.12 · forest + CHP · Volkart et al. (2013) · capture **CO_2 transport / storage:** Koornneef et al. (2008) · Terlouw et al. (2021)

Foreground systems modelled with Brightway · linked to ecoinvent background suppliers

Electricity — kWh or MWh

Heat — GJ of delivered heat

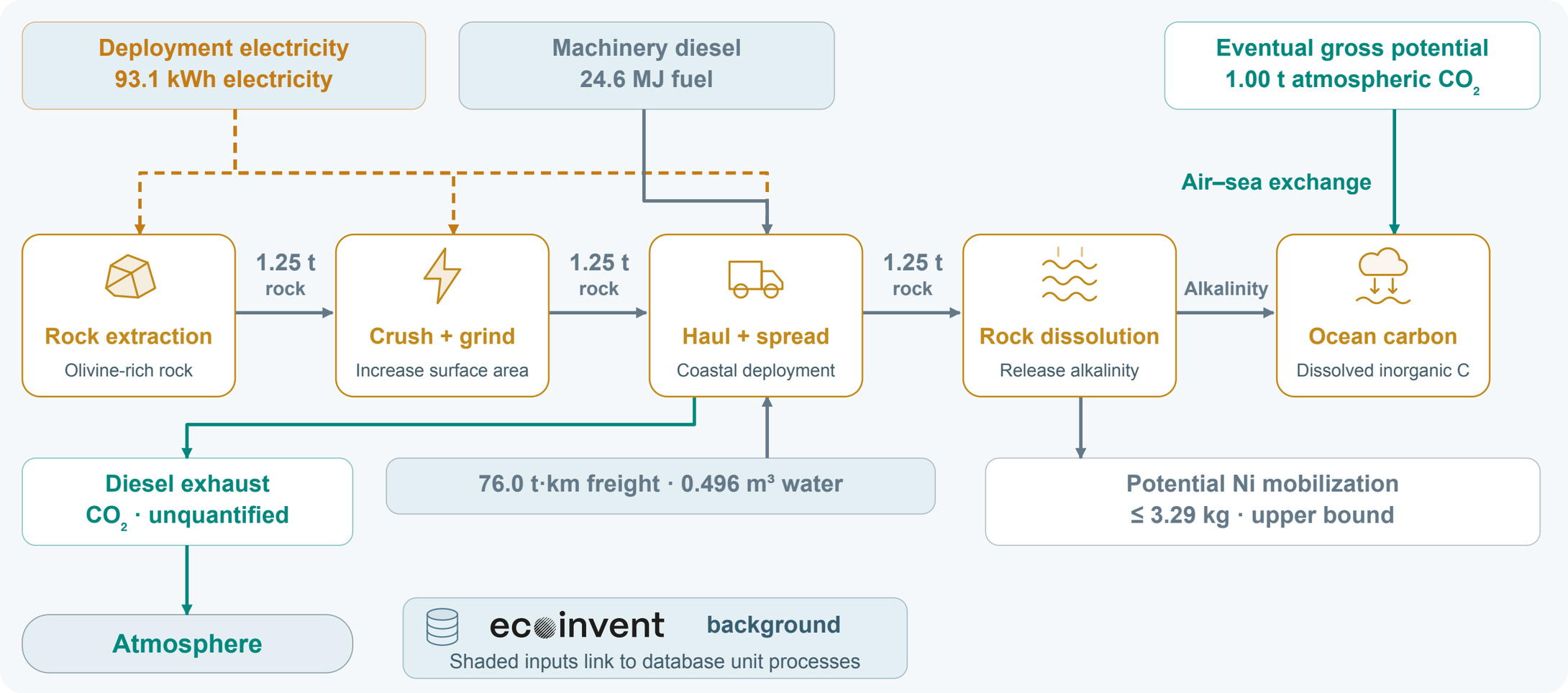
Materials + fuels — kg, t, m³ · fuel energy in MJ/GJ

CO_2 — kg or t · capture, storage, release

Follow carbon, materials and energy to storage

ERW

1.25 t rock deployed



Adapted inventory: Foteinis et al. (2023) · Supporting inventory · Table S1 · Reviewed ecoinvent 3.12 adaptation

Foreground systems modelled with Brightway · linked to ecoinvent background suppliers

Electricity — kWh or MWh

Heat — GJ of delivered heat

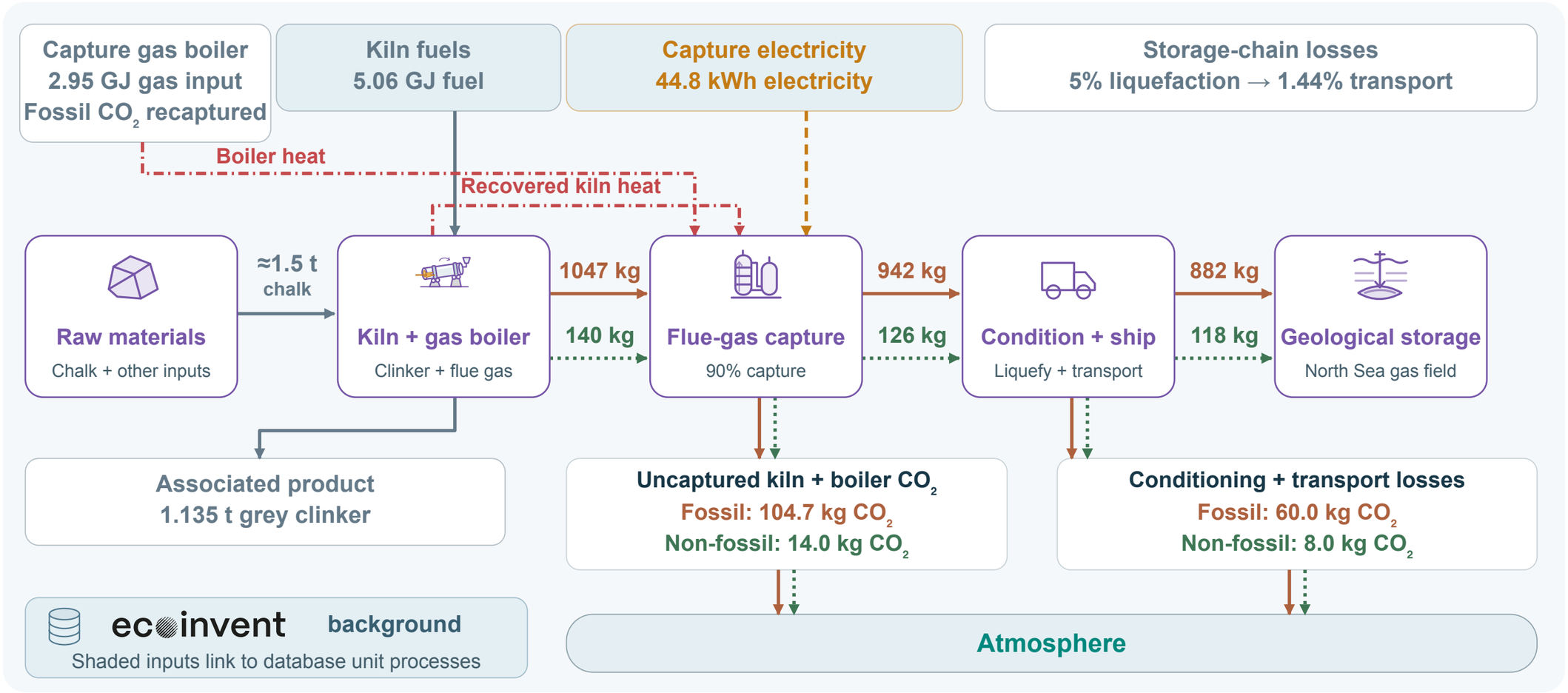
Materials + fuels — kg, t, m³ · fuel energy in MJ/GJ

CO_2 — kg or t · capture, storage, release

Follow carbon, materials and energy to storage

Cement + CCS

1 t CO_2 stored · CCS − BAU · 2025 · includes calcination + fuel + boiler CO_2



Adapted inventory: Gallego Dávila et al. (2023) · SI.2 · Reviewed cement webinar adaptation

Foreground systems modelled with Brightway · linked to ecoinvent background suppliers

Electricity
kWh or MWh

Heat
GJ of delivered heat

Materials + fuels
kg, t, m³ · fuel energy in MJ/GJ

CO_2
kg or t · capture, storage, release
Fossil CO_2 · Non-fossil CO_2

CURRENT STATIC

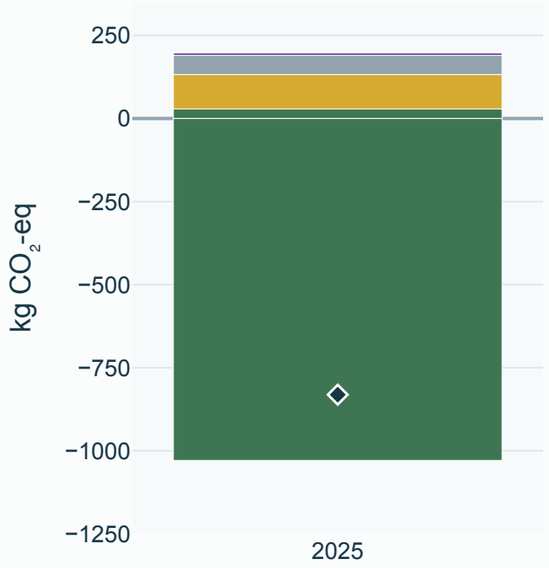
What does a present-day score tell us?

GWP100 puts each greenhouse gas on a CO_2-equivalent scale using its effect over 100 years, then adds the life-cycle contributions.

DACCS

1 t CO_2 physically stored

Northern Europe · 2025

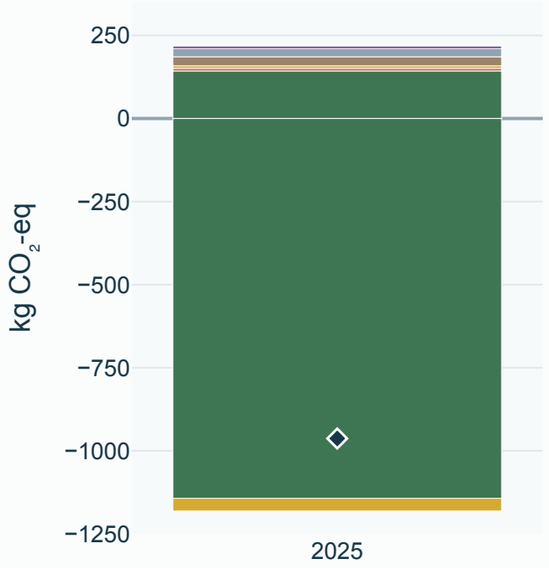


−831

BECCS

1 t CO_2 physically stored

Northern Europe · 2025

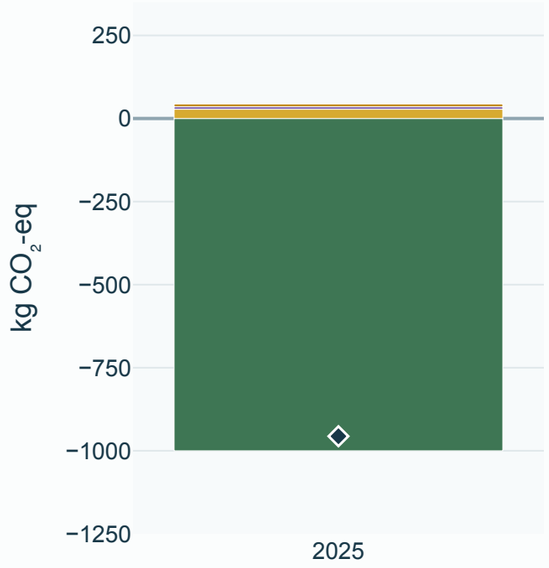


−963

ERW

1.25 t rock deployed

European coastal model · 2025

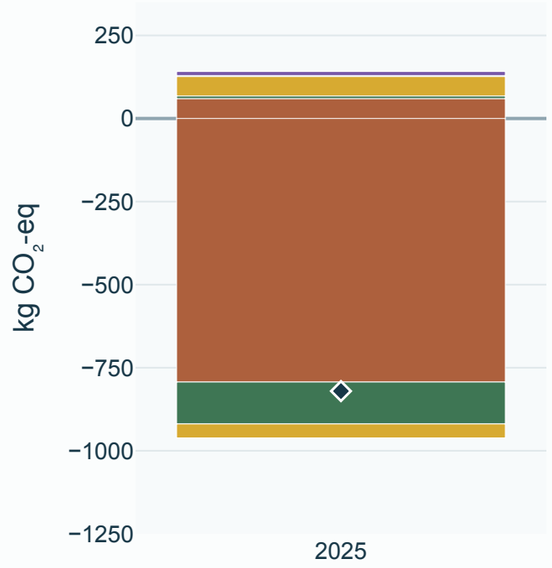


−956

Cement + CCS

1 t CO_2 stored · CCS − BAU

Aalborg, Denmark · 2025



−820

- fossil CO_2
- non-fossil CO_2
- Other direct effects
- Electricity + heat
- Biomass supply
- Materials + equipment
- Transport + storage
- Mining + spreading
- ◆ **Net GWP100**

DISCUSSION PAUSE 1 · 3 MINUTES

Would today’s supply chain describe a future project?

DACCS · BECCS · ERW · Cement + CCS

THINK · SHARE · DISCUSS

A project starts in 2035 and operates for twenty years. Which assumptions in today’s snapshot would you question?



DACCS

Electricity + heat



BECCS

Biomass + land



ERW

Grinding + transport

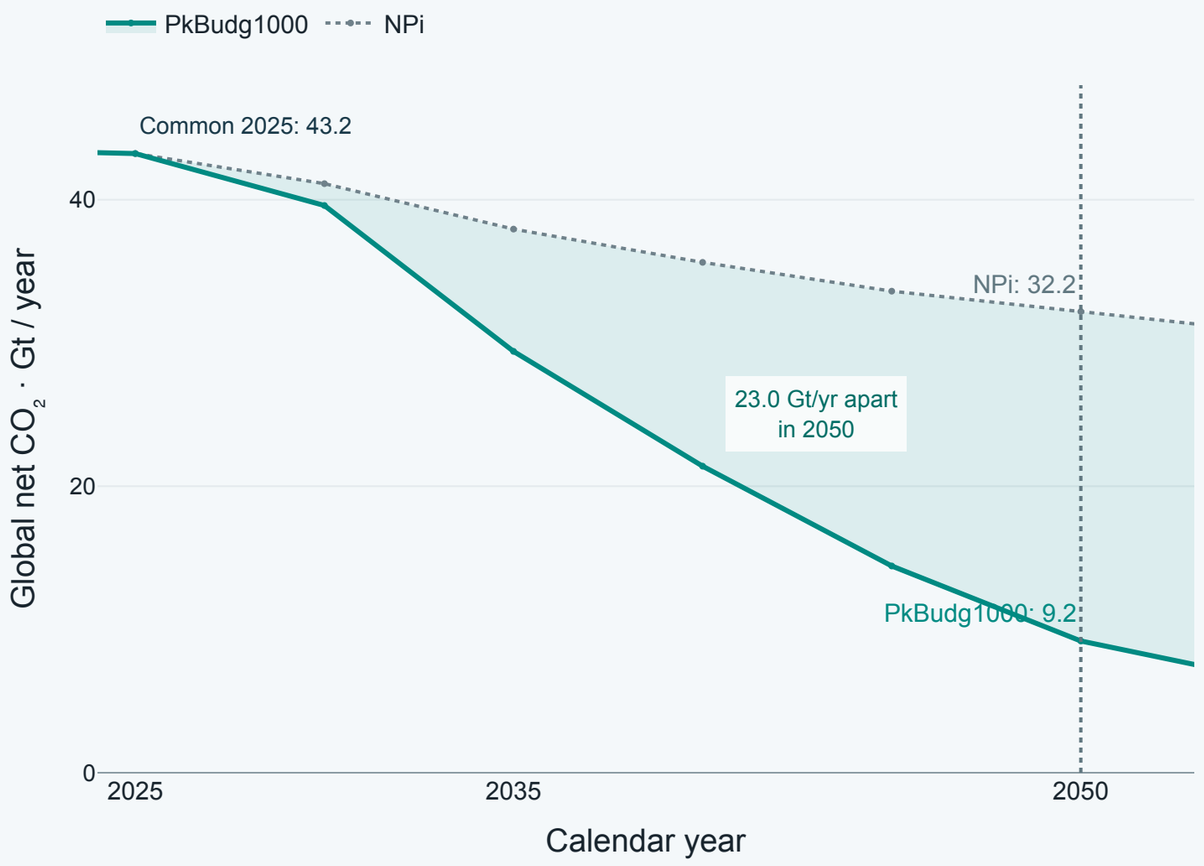


Cement + CCS

Kiln + capture energy

The same year can mean very different supply systems

SAME YEAR · DIFFERENT FUTURE SYSTEMS



2050 · TWO CONDITIONAL WORLDS

Policy changes investment choices

NPi

National policies implemented

32.2 Gt CO_2 / year

26% below the common 2025 level

PkBudg1000

Carbon-budget constraint

9.2 Gt CO_2 / year

79% below the common 2025 level

Power · heat · fuels · industrial routes

The regional and sector-specific changes determine which suppliers a future project uses.

The global curve provides context; it is not an electricity emission factor.

DACCS

Energy to run capture

Cleaner power and heat can reduce operating burdens.

BECCS

The displaced energy system

Cleaner alternatives can shrink electricity and heat credits.

ERW

Rock preparation + transport

Grinding power and haulage technologies change.

Cement + CCS

Capture utilities + kiln fuels

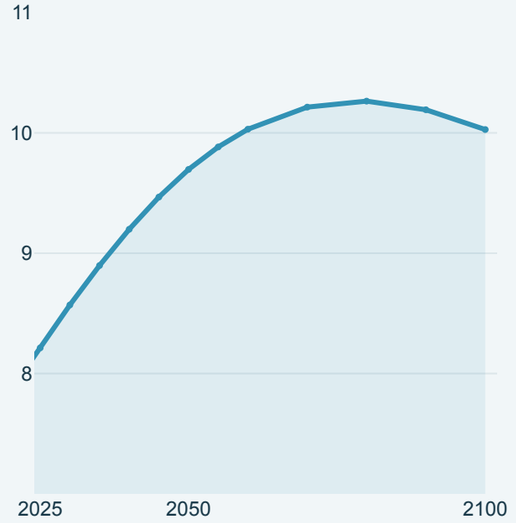
Fuel mix, heat supply and capture design evolve.

IAMs connect economies, land, resources and climate

1 · Set assumptions

Population + income
Demand for energy and services

World population · SSP2
Billion people · shared by both pathways



Technology + resources
Costs, learning and available options

Policy constraints
Taxes, targets and carbon budgets

2 · Link sectors and constraints

Economy, demand + investment
Services · consumption · trade

Energy + industry
Power · fuels · materials

Land, food + forests
Farming · bioenergy · land use

Transport + buildings
Mobility · housing · infrastructure

Water + resources
Availability · competing uses

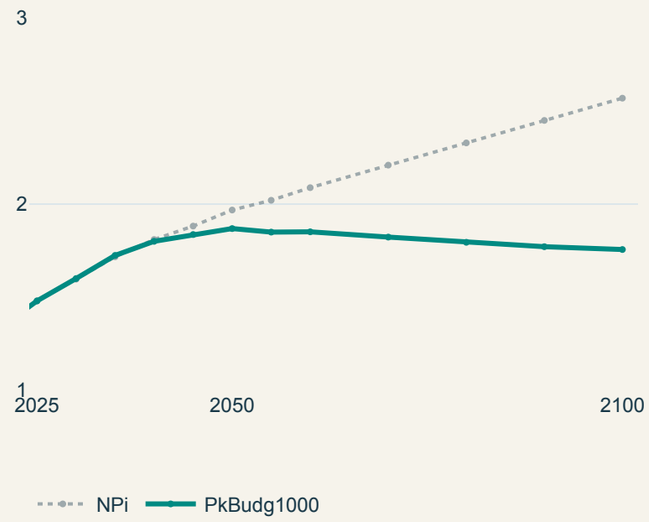
Emissions, climate + environment
Carbon cycle · air · water · ecosystems

Competing demands, shared constraints
Energy, land, food and materials interact; coverage varies by model.

IMAGE framework · human and Earth systems

3 · Read conditional outputs

Scenario climate background
Global warming · °C above 1850–1900



Stored MAGICC7 medians

Technology and fuel mixes
Power, heat, transport and industry

Land, water + emissions
Regional production and environmental pressures

→ **Evidence for future LCA suppliers**

REMIND–MAgPIE
PIK · MAgPIE mark

IMAGE
PBL · team mark

MESSAGEix–GLOBIOM
IIASA · MESSAGEix mark

GCAM
JGCRI · UMD team mark

AIM/CGE
AIM model family

WITCH
WITCH · RFF-CMCC mark

Scenarios combine development stories and climate assumptions

Societal storyline — How might society develop? + **IAM model** — How are systems represented? + **Climate assumptions** — Policies or forcing constraint → **Conditional pathways** — Energy, land, production, emissions

SOCIETY · FIVE DEVELOPMENT STORIES

Shared Socioeconomic Pathways

SSP1 **Sustainability**
Cooperation, resource efficiency, less inequality.

SSP2 **Middle of the road**
Uneven progress; development follows past trends.

SSP3 **Regional rivalry**
Fragmentation, weak cooperation, slow development.

SSP4 **Inequality**
Large gaps in wealth, opportunity and capabilities.

SSP5 **Fossil-fuelled development**
Rapid growth and energy-intensive fossil use.

MODEL · COMBINE THE ASSUMPTIONS

One SSP, several climate pathways

SSP1 → SSP1-1.9, SSP1-2.6

The IAM translates socioeconomic and climate assumptions into regional, sector-specific trajectories.

In this lecture

SSP2 + REMIND-EU

~2.6 °C in 2100
Implemented national policies

<2 °C in 2100
Carbon-budget constraint

Median GMST · above 1850–1900

CLIMATE · FIVE AR6 EXAMPLES

Illustrative emissions pathways

Numeric suffix ≈ forcing in 2100 · W/m²

SSP1-1.9 **Very low emissions**
Rapid cuts; net-zero CO_2 around mid-century.

SSP1-2.6 **Low emissions**
Strong cuts; net-zero CO_2 later in the century.

SSP2-4.5 **Intermediate emissions**
CO_2 stays near today's level, then declines.

SSP3-7.0 **High emissions**
Emissions keep rising through much of the century.

SSP5-8.5 **Very high emissions**
Fossil-intensive growth; CO_2 roughly doubles by 2050.

IPCC · Intergovernmental Panel on Climate Change

Research teams develop scenarios → IPCC assesses the evidence

WG I Climate response

WG II Impacts + adaptation

WG III Mitigation pathways

IPCC AR6 · SSP framework · AR6 illustrative scenarios · IPCC assessment role

Trace a scenario change into the LCA result

premise documentation ↗

Open scenario explorer ↗

REMIND-EU scenarios
Integrated assessment model · SSP2

premise
Updates mixes, efficiency and emissions

→

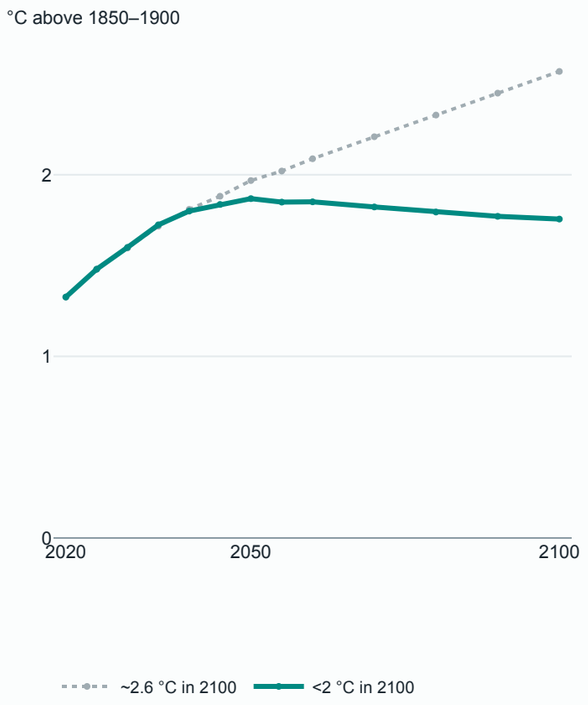
Prospective ecoinvent
Northern Europe (ENC) · 2025–2050

IAM scenarios and Northern European supply mixes

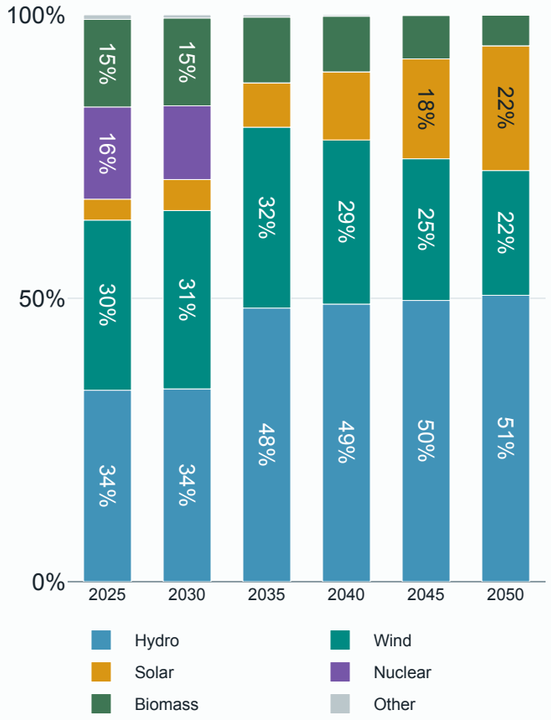
Mixes: <2 °C in 2100 · Northern Europe (ENC)

GMST

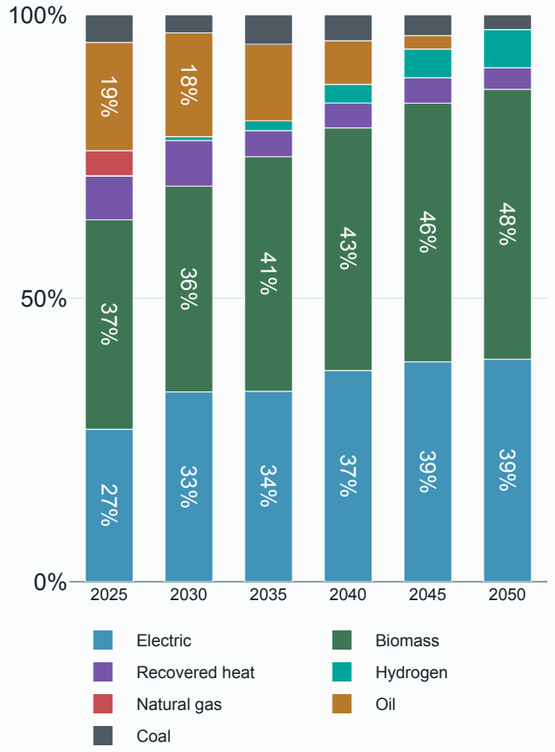
World



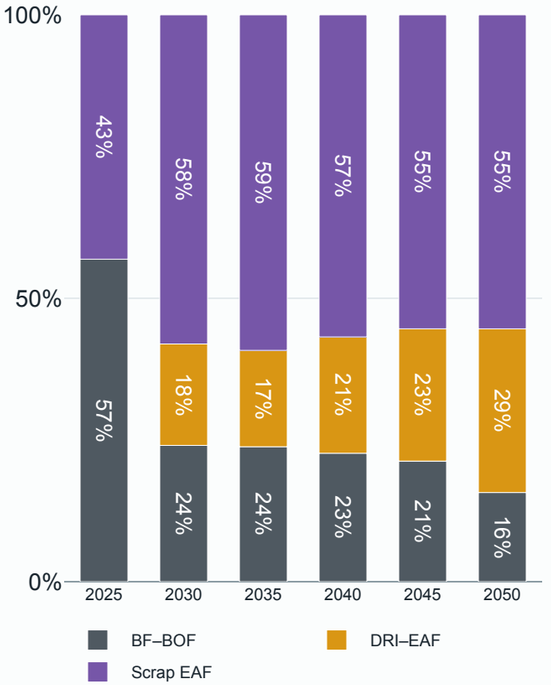
Electricity



Heat supply



Steel



Future supply chains change both burdens and credits

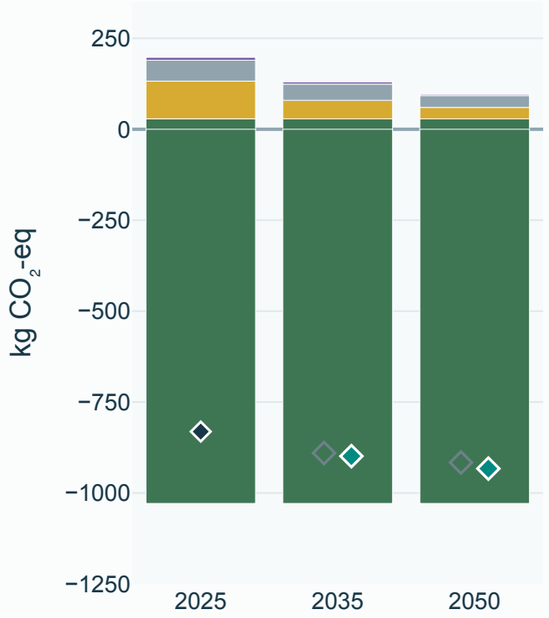
Future bars: <2 °C in 2100 (default)

◇ ~2.6 °C in 2100 net ◆ <2 °C in 2100 net ◆ 2025 reference

DACCS

1 t CO_2 physically stored

Northern Europe · 2025/2035/2050

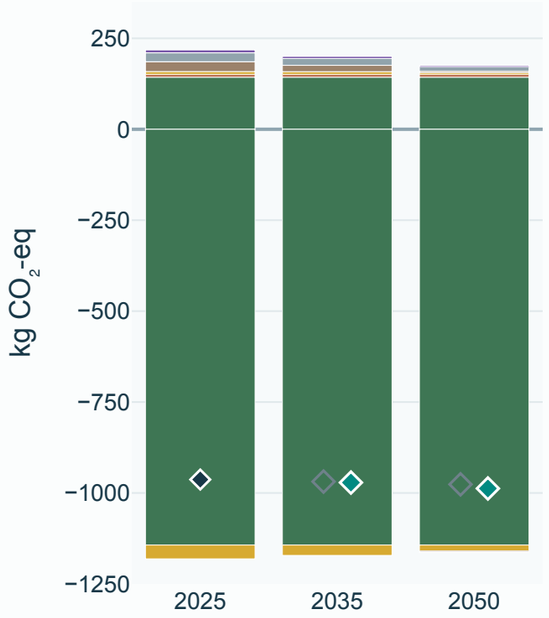


2050: −933

BECCS

1 t CO_2 physically stored

Northern Europe · 2025/2035/2050

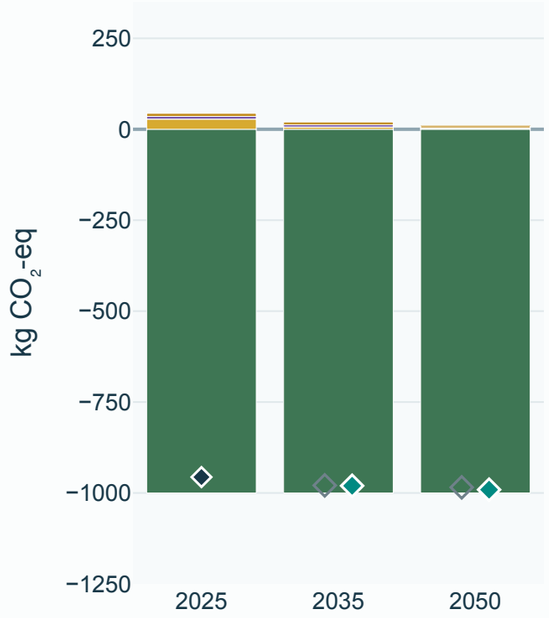


2050: −988

ERW

1.25 t rock deployed

European coastal model · 2025/2035/2050

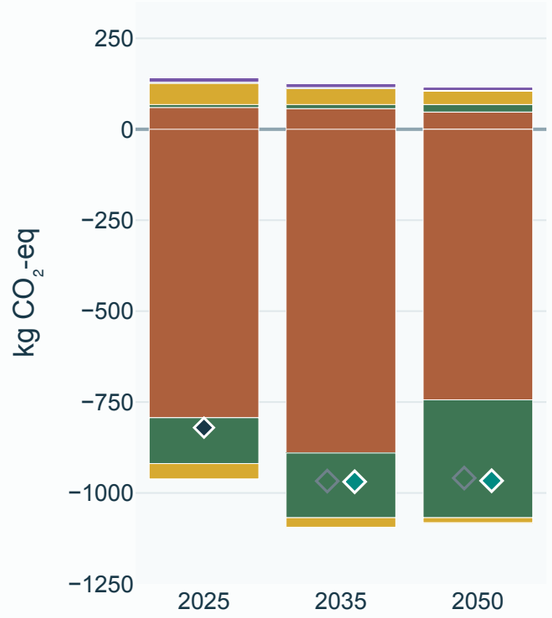


2050: −991

Cement + CCS

1 t CO_2 stored · CCS − BAU

Aalborg, Denmark · 2025/2035/2050



2050: −966

fossil CO_2 · non-fossil CO_2 · Other direct effects · Electricity + heat · Biomass supply

Materials + equipment · Transport + storage · Mining + spreading

DISCUSSION PAUSE 2 · 3 MINUTES

Is one future snapshot enough for a long-lived system?

DACCS · BECCS · ERW · Cement + CCS

THINK · SHARE · DISCUSS

Two projects have the same aggregate score. One removes CO_2 today; the other removes it fifty years later. What is missing?

The background economy changes, too

Electricity + heat mixes can evolve during operation.

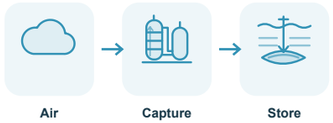
Early operation

Electricity + heat suppliers

→

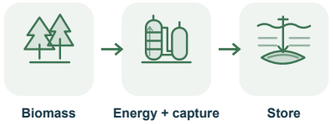
Later operation

Year-specific suppliers



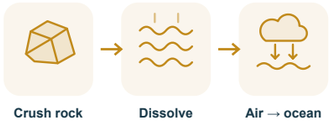
DACCS

Capture during operation



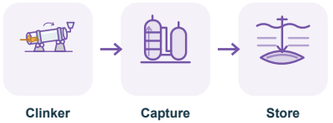
BECCS

Forest regrowth



ERW

Dissolution + air–sea exchange



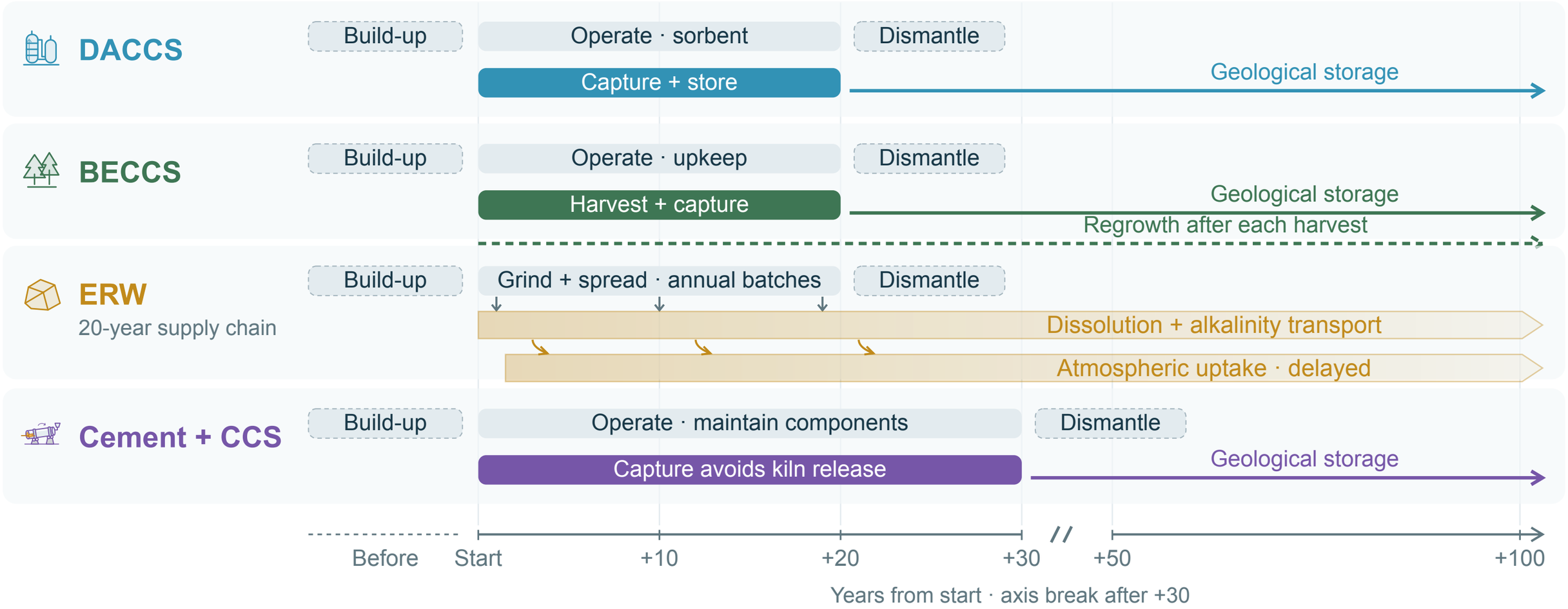
Cement + CCS

Construction + avoided emissions

Equipment and carbon follow different clocks

Equipment + recurring inputs Carbon pathways

Equipment stages + ERW arrows schematic



From scenarios to dated supply chains

01 · Assemble the time-aware model

IAM scenarios
REMIND-EU
Pathway + future years

+

ecoinvent
3.12 · cut-off inventories

↓

premise

premise: scenario-updated LCA data
Transform suppliers · export scenario years

↓

Scenario data package
3D stacks: two inventory dimensions + year

Technosphere A



activity × product × year

Biosphere B



activity × flow × year

+ Encode temporal distributions
Analyst's foreground: build · operate · regrow / dissolve

↓

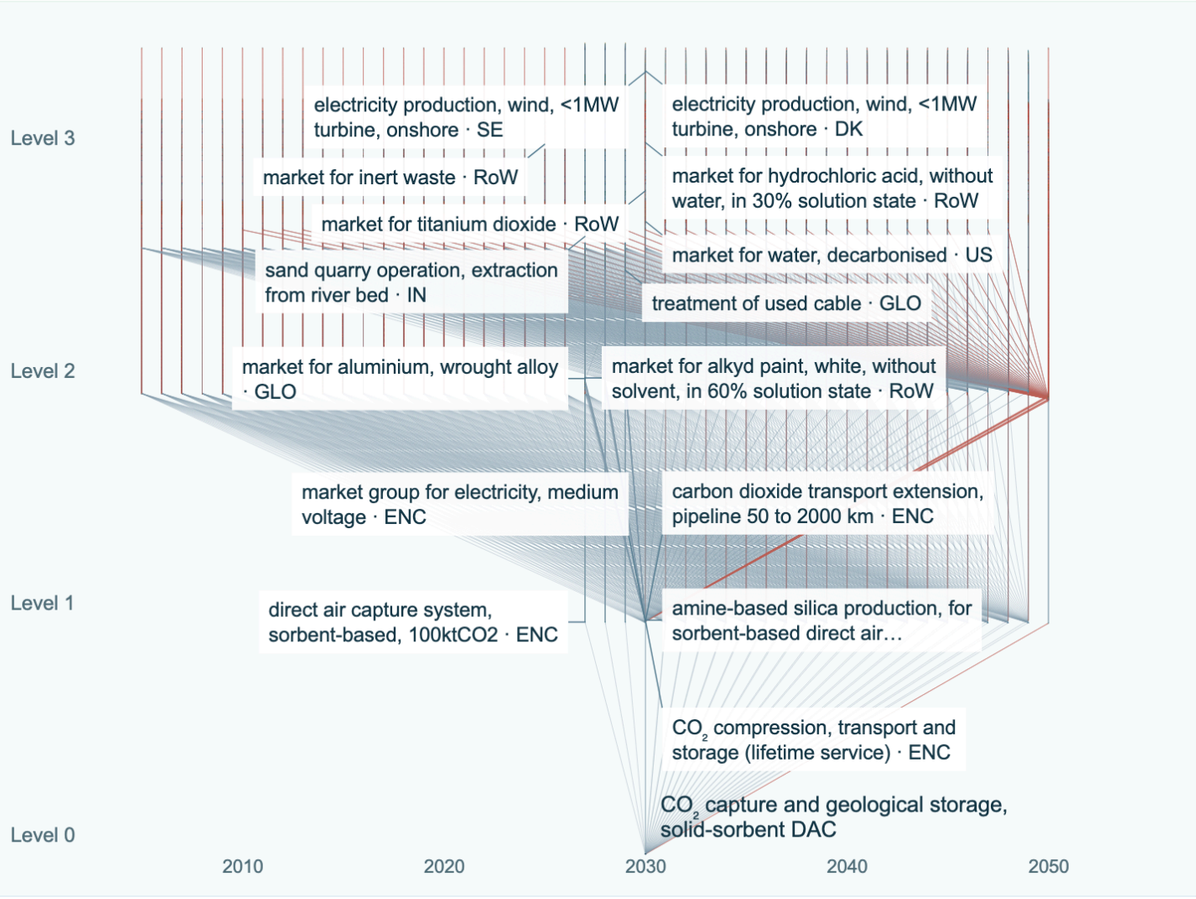
TRAILS: route in time → solve by year
Interpolate backgrounds · retain the remaining supply chain

02 · Follow the temporal routes

Actual TRAILS output · functional unit at level 0 → level 4

DACCS Open full graph ↗

Level 0 · 1 t CO_2 physically stored



TIME-EXPLICIT LCA

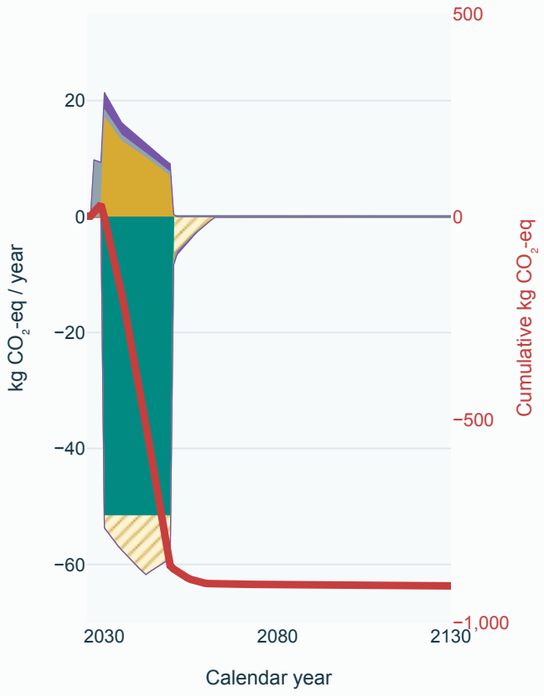
What drives the GWP score, and when?

Whole-project activity per 1 t CO_2 stored

DACCS

1 t CO_2 stored · whole project

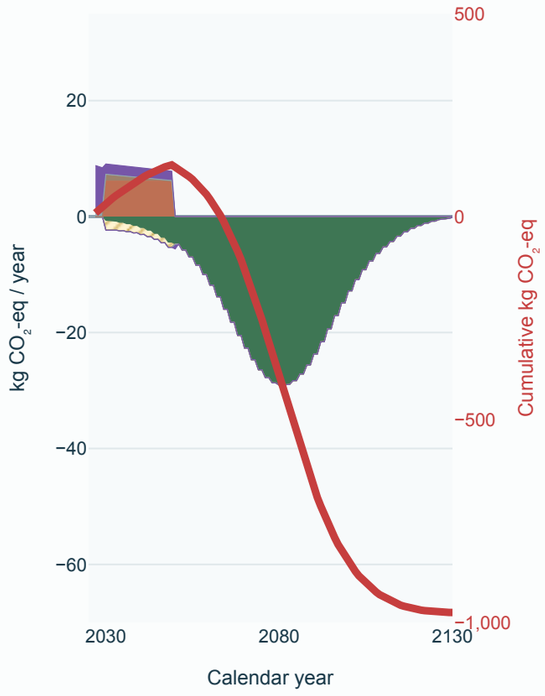
2030 cohort



BECCS

1 t CO_2 stored · whole project

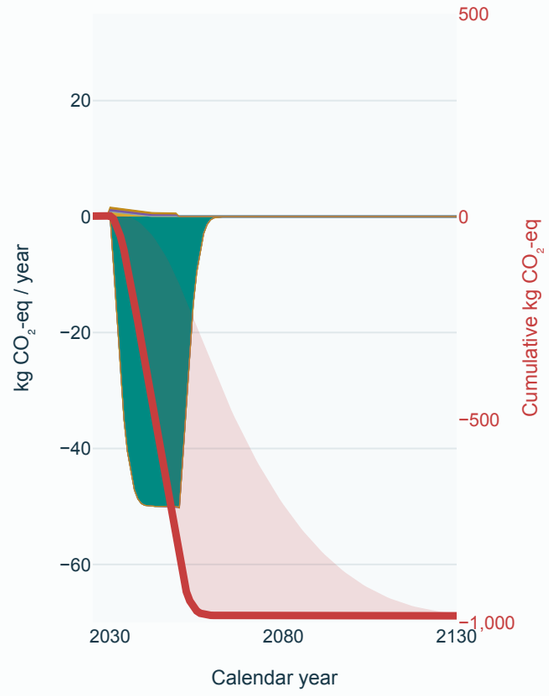
2030 cohort



ERW

1 t eventual uptake · modelled

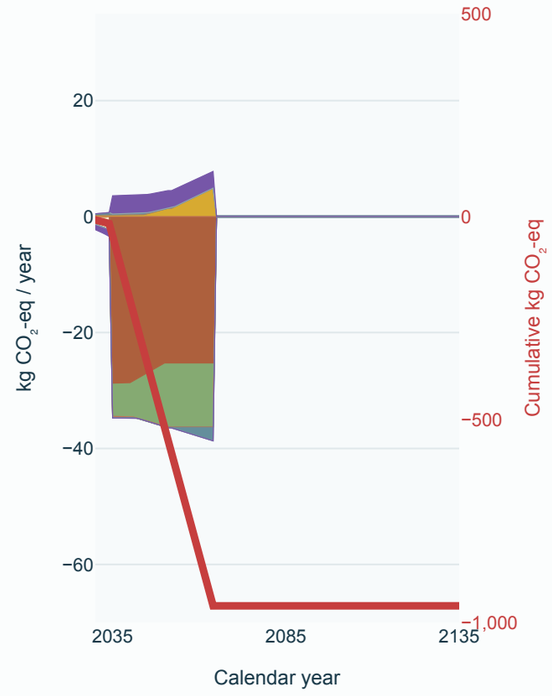
2030–2049 · illustrative 10 y / 1 y



Cement + CCS

1 t CO_2 stored · whole project

2035 · CCS − BAU · all CO_2



- CO_2 uptake
- Biogenic uptake
- Avoided fossil CO_2
- Avoided biogenic CO_2
- Other foreground effects
- Electricity + heat (+)
- Electricity + heat (−)
- Fuel + biomass supply
- Capture operation + heat credit
- Materials + equipment
- Transport + storage
- Mining + spreading
- ERW timing scenarios
- **Cumulative GWP100 · right axis**

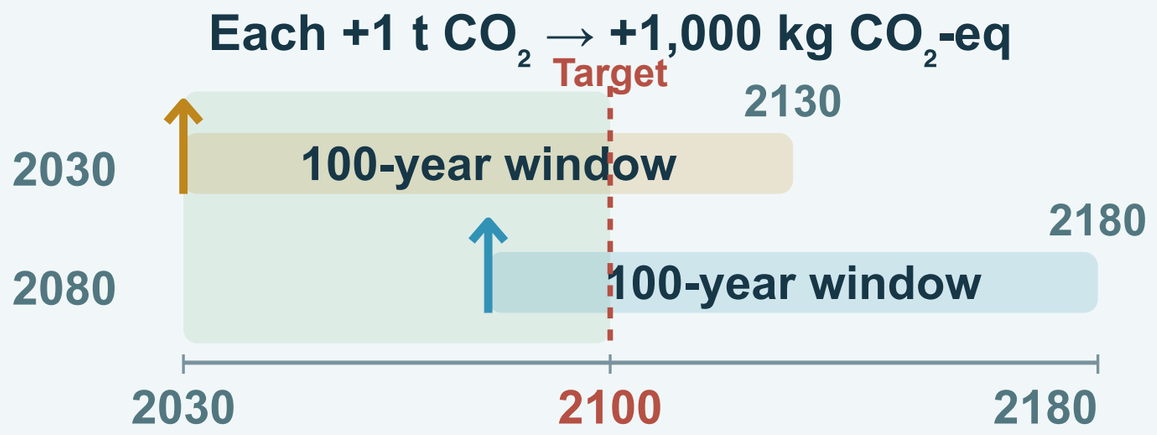
Same GWP100. Same climate outcome?

THINK · COMPARE · DISCUSS

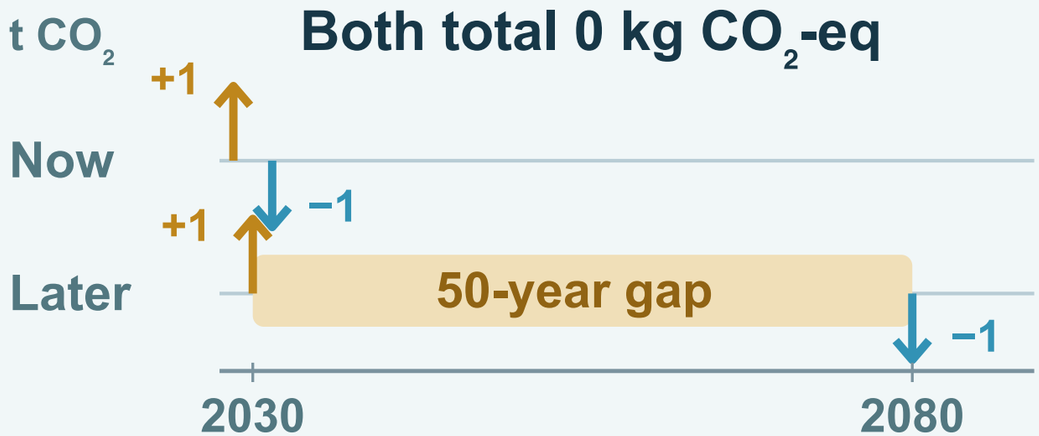
IPCC AR6 §7.6 Levasseur et al. (2010) Reisinger et al. (2011)

Do dated GWP100 scores answer our climate question?

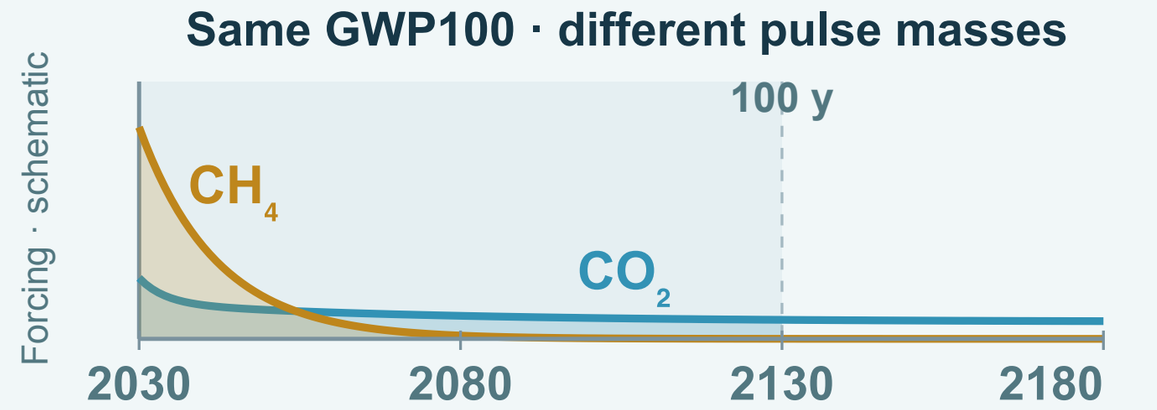
01 Same score, different assessment window?



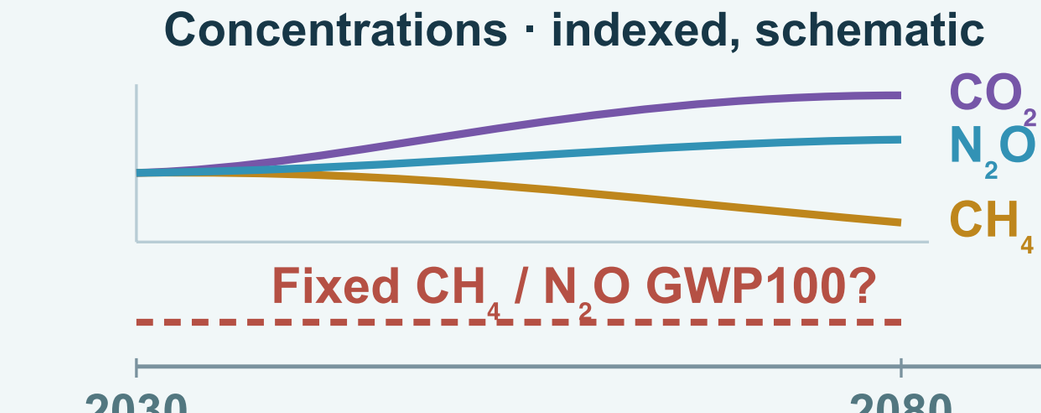
02 Zero total, same climate history?



03 Same CO_2-eq, different persistence?



04 Same factors in a changing atmosphere?



CLIMATE RESPONSE

Later removal does not erase the earlier warming

DACCS · BECCS · ERW · Cement + CCS

2030 · EMIT

+1 t CO_2

A pulse enters the atmosphere

WHILE WE WAIT

A warming influence persists

The atmosphere responds before removal

REMOVE AT THE SELECTED DATE

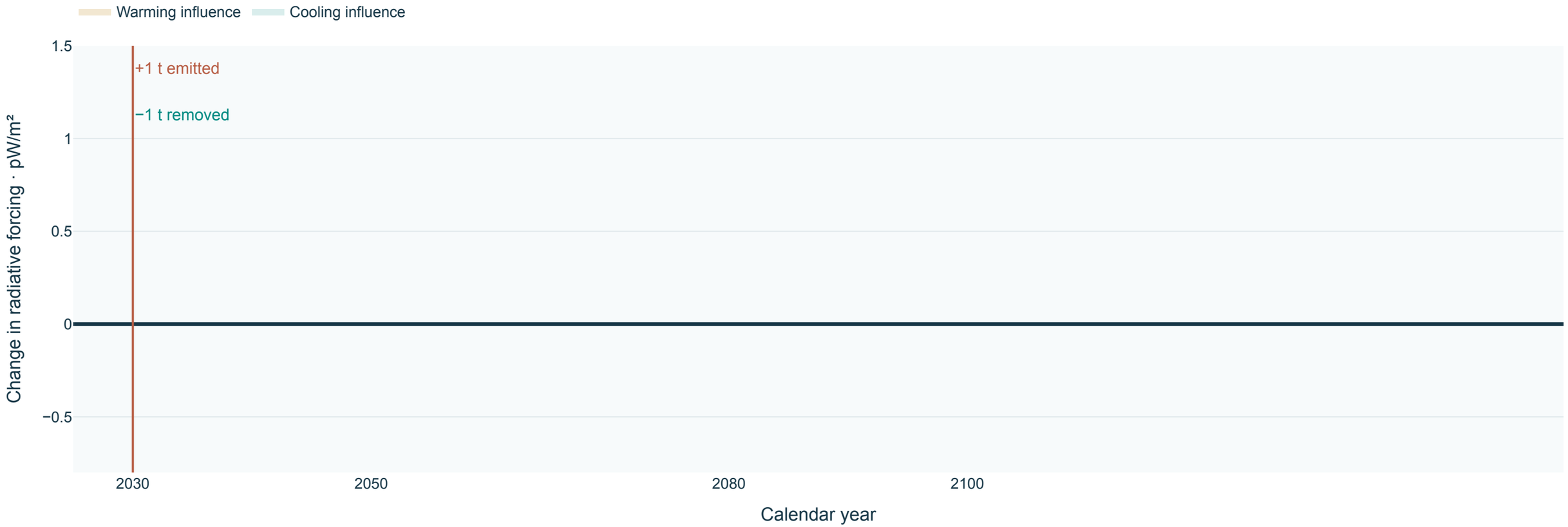
−1 t CO_2

Later uptake changes the subsequent response

Aggregate GWP100: 0 kg CO_2-eq

+1 t minus 1 t, regardless of the delay

Removal: 2030 · same year



CLIMATE RESPONSE

Follow all four climate responses through time



premise → TRAILS → FaIR

Calendar window: 2030–2050

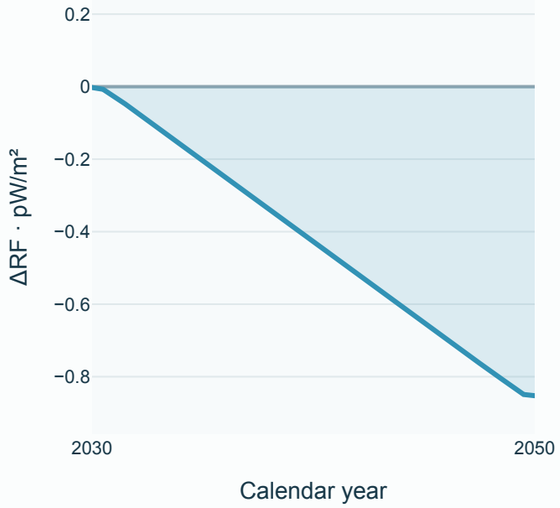
DACCS

1 t CO_2 physically stored

2030 cohort

Integrated forcing

−8.55 pW·year/m²



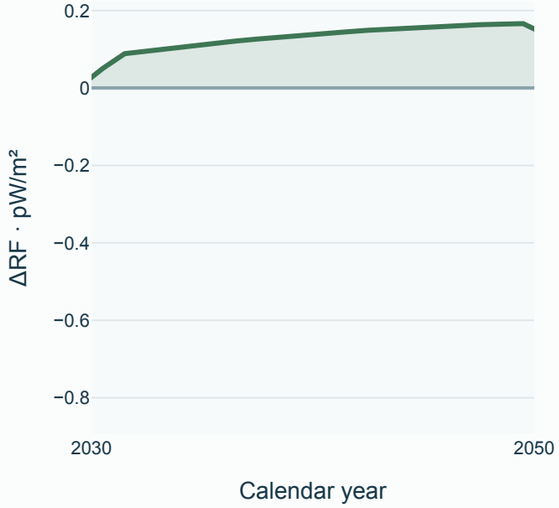
BECCS

1 t CO_2 physically stored

2030 cohort

Integrated forcing

+2.63 pW·year/m²



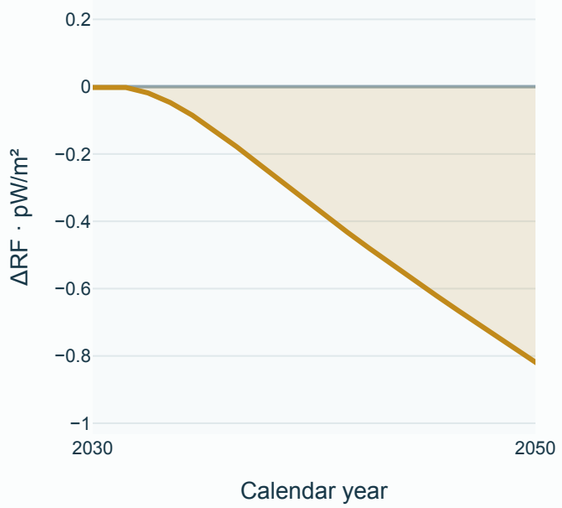
ERW

1.25 t rock deployed

2030–2049 · illustrative 10 y / 1 y

Integrated forcing

−7.25 pW·year/m²



Cement + CCS

1 t CO_2 stored · CCS − BAU

2035 · CCS − BAU

● Fossil / mineral ● Non-fossil
Combined response

Integrated forcing

−4.18 pW·year/m²

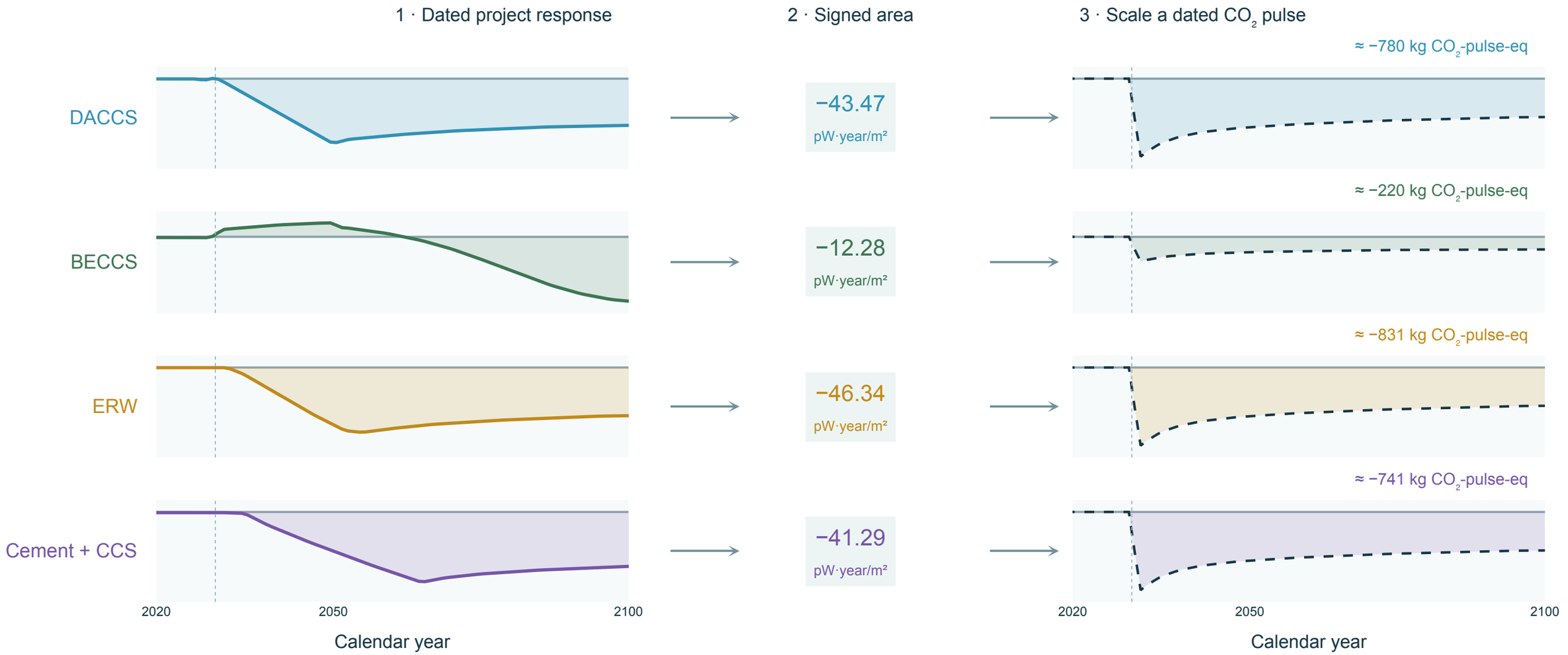


CO_2-pulse-eq: match effects over a declared window

Equal signed areas, different trajectories

Within each row: same background, calendar window and vertical scale.

Window 2020–2100 | CO_2 reference pulse in 2030 | Illustrative matches of displayed curves



TRAILS · fixed-window CO_2-pulse equivalence · Levasseur et al. (2010)

CLIMATE RESPONSE

CO_2-pulse-eq: match effects over a declared window

Match each project's signed area with a dated CO_2 pulse

Illustrative curve matches · native climate background for each case

Start 2020 · CO_2 pulse 2030 · end 2100

Project response Area-matched CO_2 pulse

2020–2100 · pulse 2030

DACCS

1 t CO_2 physically stored

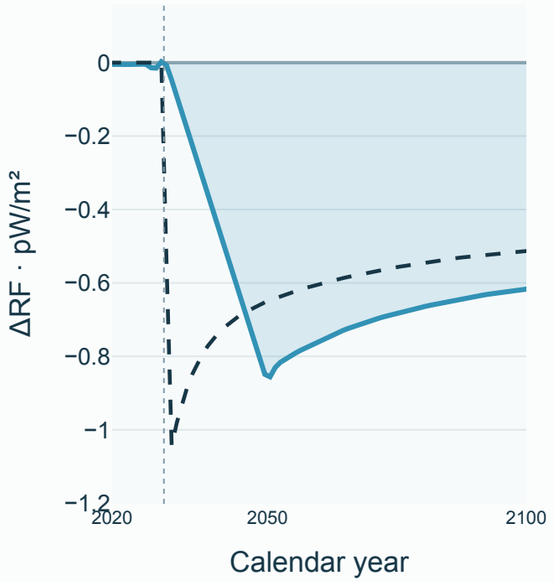
2030 cohort

Area-matched reference pulse

≈ −780

kg CO_2-pulse-eq

Both areas: −43.47 pW·year/m²



BECCS

1 t CO_2 physically stored

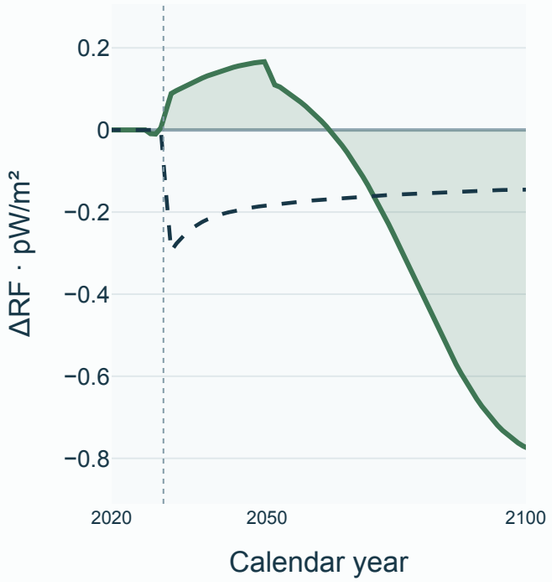
2030 cohort

Area-matched reference pulse

≈ −220

kg CO_2-pulse-eq

Both areas: −12.28 pW·year/m²



ERW

1.25 t rock deployed

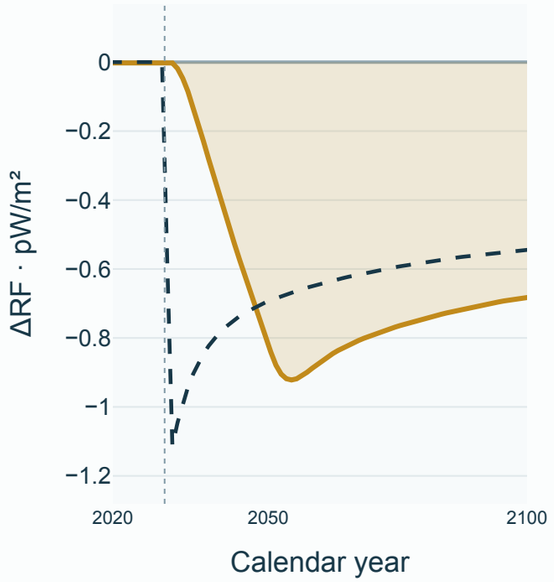
2030–2049 campaign

Area-matched reference pulse

≈ −831

kg CO_2-pulse-eq

Both areas: −46.34 pW·year/m²



Cement + CCS

1 t CO_2 stored · CCS − BAU

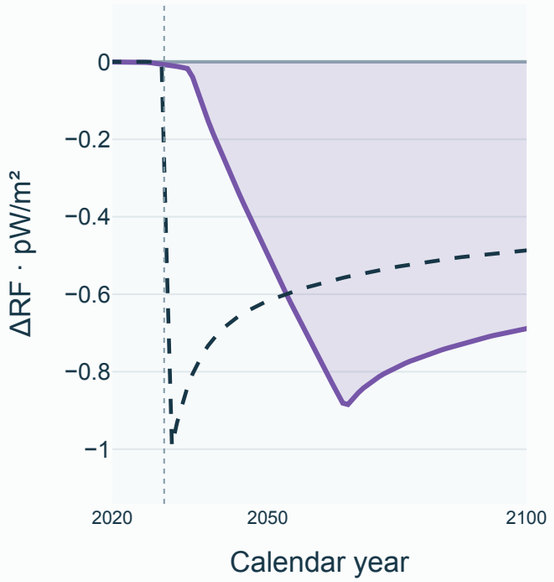
2035 · CCS − BAU

Area-matched reference pulse

≈ −741

kg CO_2-pulse-eq

Both areas: −41.29 pW·year/m²



TRAILS · fixed-window CO_2-pulse equivalence · Levasseur et al. (2010)

TOOLING AND SOURCES

From a project question to a dated climate result

DACCS · BECCS · ERW · Cement + CCS

Analyst choices

Service
What does the system deliver?

Reference
What happens without it?

Timing evidence
When do the flows occur?

Climate question
Background · window · pulse

01 Frame



Foreground + reference

Specify the service and the system without the intervention.

Study definition

02 Transform

ecoinvent
+ IAM scenarios
premise

ecoinvent · IAMs · premise

Translate scenario pathways into changing life-cycle suppliers.

Future supplier data

03 Date + solve



TRAILS

Route temporal exchanges and solve the year-specific supply chains.

GHG emissions by year

04 Simulate

FaIR



FaIR

Compare the same climate background with and without the project.

Forcing + temperature

05 Interpret



Trajectory + CO_2-pulse-eq

Report the response curve together with a declared comparison window.

A dated climate result

TOOLING AND SOURCES

Where to start with your own study

DACCS · BECCS · ERW · Cement + CCS

01 Frame and model your system

Define the system boundaries and the timing question your assessment must answer.

First step Start modelling your system and its boundaries with Brightway; specify the service, reference, emission dates and climate window.

Brightway · foreground modelling · Dynamic LCA · Levasseur 2010 · IPCC · emission metrics

02 Build future suppliers

Turn an IAM pathway into prospective life-cycle inventory data.

First step Choose an ecoinvent system model, then run premise's first scenario example.

ecoinvent · system models · premise · first scenario

03 Calculate the timing effects

Route dated exchanges, then inspect forcing and pulse equivalents.

First step Start with the TRAILS quickstart; use FaIR's calibrated ensemble example for climate modelling.

TRAILS · quickstart · FaIR · calibrated example

04 Adapt an ERW inventory

Separate the source inventory from site-dependent uptake assumptions.

First step Open the reviewed workbook, then check grinding, grain size and uptake timing for your site.

Reviewed inventory + changes · Source inventory · Table S1 · Marine weathering · Geerts 2025

Revisit a worked example IAM scenarios · CDR + time · Cement + CCS · Temporal LCA

Life-cycle scores for the four options

Native inventories and climate backgrounds · 1 t stored, or eventual ERW uptake potential

Option	GWP100 · kg CO_2-eq			CO_2-pulse-eq	Gross CO_2 needed
	Current 2025	Future 2035	Time-explicit Operation period	kg · curve match 2020–2100 · ref. 2030	t captured / ERW uptake Match −1 t in 2030 RF area · 2020–2100
DACCS	−831	−899	−911 2030–2049	≈ −780	≈ 1.32
BECCS	−963	−971	−976 2030–2049	≈ −220	≈ 4.67
ERW	−956	−980	−984 2030–2049	≈ −831	≈ 1.20
Cement + CCS	−820	−969	−959 2035–2064	≈ −741	≈ 1.44

- **Prospective LCA** captures changing production conditions.
- **Time-explicit LCA** reveals when emissions and climate benefits occur.
- **CO_2-pulse-eq** summarizes an effect over a declared window; keep the trajectory alongside it.

Seoul National University · 30 October 2026

Graduate LCA seminar

With thanks to Jooyoung Park and the course participants.

Thank you

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Horizon Europe · UPTAKE
European Union · Grant agreement 101081521

The conventional and prospective matrix equations

$$A\,x = f \rightarrow x = A^{-1}\,f$$

$$s = c^{T}\,B\,A^{-1}\,f$$

$$s(\sigma, y) = c^{T}\,B(\sigma, y)\,A(\sigma, y)^{-1}\,f(y)$$

A, B

Technosphere and biosphere exchanges

f, x

Final demand and process scaling

c, s

Characterization factors and impact score

Time-resolved inventory and dynamic characterization

$$\Sigma_j \Sigma_u A_{ij}(t, u)\, x_j(u) = f_i(t)$$

$$I_g(t) = \Sigma_j \Sigma_u B_{gj}(t, u)\, x_j(u)$$

$$\text{Static factors: } S = \Sigma_t \Sigma_g I_g(t) \cdot GWP_{100,g}$$

$$\text{Dynamic response: } \Delta R(\tau) \approx \Sigma_{t \le \tau} \Sigma_g I_g(t) \cdot r_g(\tau, t)$$

$$\text{Common window: } J(H) = \int_{t_0}^{H} \Delta R(\tau)\, d\tau$$

t, u

Exchange-event year and year of activity operation

A_{ij}

Dated product exchanges per unit of activity; balance every product and year

B_{gj}

Dated elementary exchanges; retain gas or flow identity

t, τ, H

Event year, response year and common calendar horizon

r_g

Gas-specific marginal response kernel in a defined background state

ΔR

Forcing or temperature perturbation; FaIR can simulate this directly

What is reused, calculated and assumed?

01

Reused

IAM workshop visuals; SETAC/lca-time CDR results; reviewed cement webinar.

02

Calculated

ERW current/future snapshots, depth-one dated LCI and FaIR response.

03

Assumed

Coastal uptake kinetics and retained proxies; source-specific references and cohorts.

CO_2-pulse-equivalence: definition and reporting

$$MCO_2\text{-pulse-eq} = Mref \times \int \Delta RFLCA(t)\,dt \,/\, \int \Delta RFpulse(t)\,dt$$

Same limits

Integrate both responses from t_0 to t_1

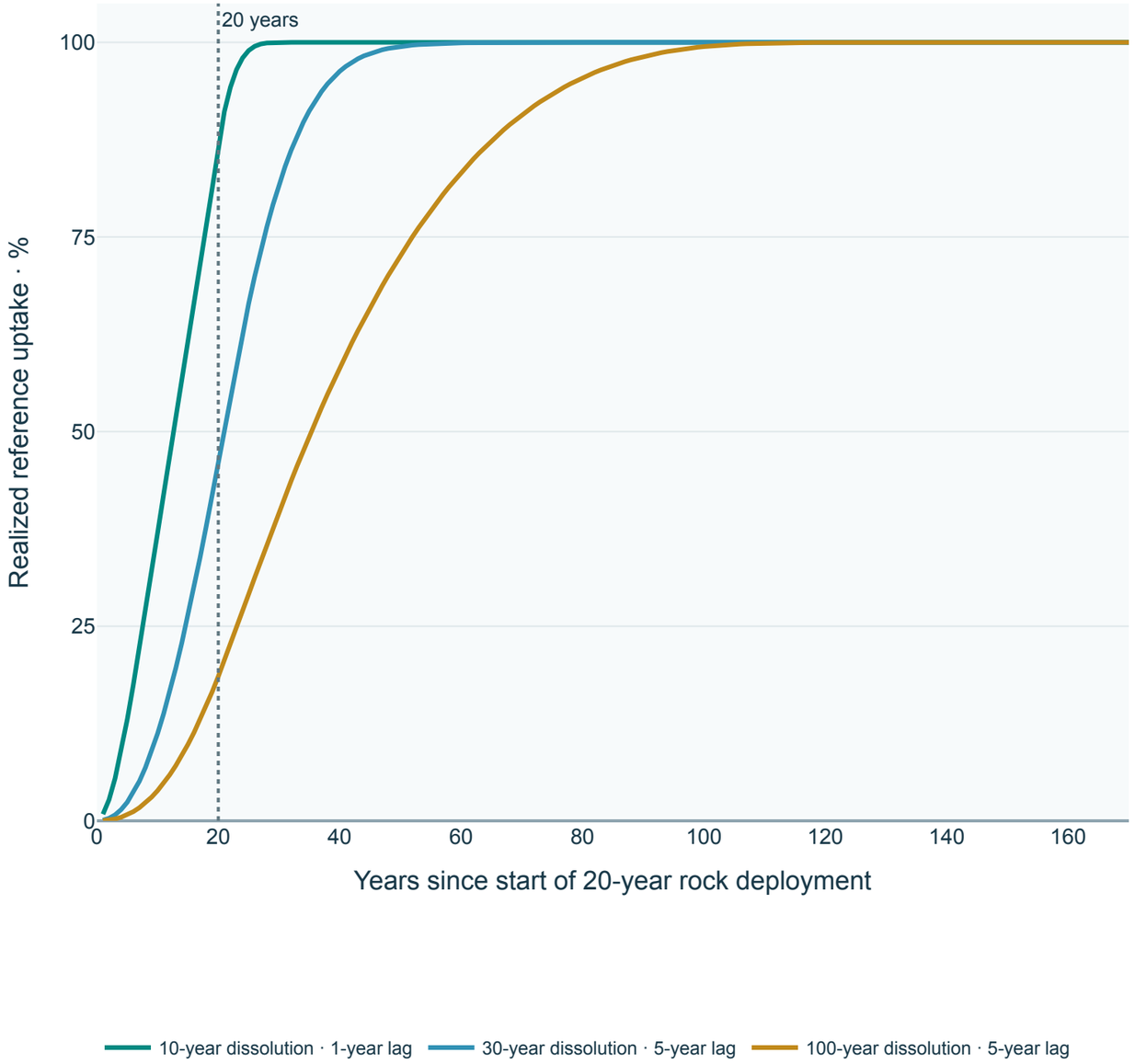
Reference

CO_2 pulse mass Mref at declared year tp

Ensemble

Compute a ratio per configuration, then summarize

ERW uptake is not bounded by a plant lifetime



Evidence and assumptions are separate

- Mineral dissolution can take years to centuries in coastal sediments.
- Atmospheric uptake follows alkalinity generation and site-dependent air–sea exchange.
- These 10 / 30 / 100-year curves are sensitivities, not a field-calibrated probability range.
- The same total rock is spread over twenty annual batches. Grain-size changes would also require new grinding inputs.

Illustrative clock	Uptake by year 20
10-year dissolution · 1-year lag	87%
30-year dissolution · 5-year lag	46%
100-year dissolution · 5-year lag	19%

Geerts et al. (2025) · He & Tyka (2023)

How root-activity contributions reconcile to the score

One root, its upstream chain

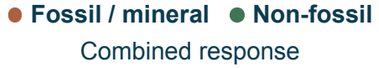
Attribute each upstream contribution to the foreground activity that demanded it. Direct foreground exchanges remain assigned to their own root.

Keep the sign before grouping

At each year, group positive and negative root scores separately by activity type. Then add them to recover the net score. Never add stage and flow decompositions together.

kg CO_2-eq per native service

Case / native service	2050 snapshot	Complete dated inventory
DACCS	−933	−911
BECCS	−988	−976
ERW	−991	−984
Cement + CCS	−966	−959



Five SSPs: the development stories behind the labels

SSPs describe development conditions before additional climate policy is imposed. They are conditional stories, not forecasts.

Storyline	People, institutions and distribution	Energy, technology and implications for LCA	Challenges: mitigation / adaptation
SSP1 **Sustainability** Taking the Green Road	Investment in education and health, falling inequality and stronger cooperation. Population peaks and declines.	Well-being and efficiency reduce material intensity. Low-carbon technologies spread; land and resource pressures still need checking.	**Low / low**
SSP2 **Middle of the Road** Historical trends continue	Moderate population growth and uneven development. Institutions improve slowly; inequality declines only gradually.	Energy intensity falls, but growing consumption raises service demand. Technology improves at a moderate pace; environmental damage persists.	**Medium / medium**
SSP3 **Regional Rivalry** A Rocky Road	National security and self-reliance displace cooperation. Weak health and education investment sustains high population growth in poorer regions.	Trade barriers, slow innovation and material-intensive production favour domestic fossil resources. Stringent mitigation is particularly difficult.	**High / high**
SSP4 **Inequality** A Road Divided	A connected, wealthy elite coexists with vulnerable, low-income groups. Opportunity, political power and adaptation capacity are highly unequal.	Advanced sectors innovate rapidly while low-tech production persists elsewhere. Mixed energy systems and local environmental protection hide large regional contrasts.	**Low / high**
SSP5 **Fossil-fuelled Development** Taking the Highway	Fast growth, integrated markets and strong health and education investment. Population peaks and declines; institutions have high adaptive capacity.	Energy-intensive lifestyles and abundant fossil fuels drive high demand. Fast innovation alone does not deliver low emissions; mitigation requires major system change.	**High / low**

The same SSP can support several climate pathways. A storyline does not determine one RCP; model feasibility and policy assumptions matter.

IPCC · SSP narratives and challenges
O'Neill et al. (2017) · SSP narratives

RCPs and emissions pathways: read the whole trajectory

The number is an approximate forcing level in 2100, in W/m² relative to pre-industrial conditions. It is not a temperature in °C.

CMIP5 · four RCPs

Representative Concentration Pathways: concentration and forcing experiments, without a unique socioeconomic story.

RCP2.6 — **Peak, then decline**
Forcing peaks near 3 W/m² before falling to about 2.6 in 2100. Strong mitigation limits cumulative emissions.

RCP4.5 — **Stabilisation at a lower level**
Emissions are constrained to approach 4.5 W/m², stabilising after 2100 without overshooting that level.

RCP6.0 — **Stabilisation at a higher level**
A higher forcing pathway approaches 6.0 W/m² after 2100. Its timing and cumulative emissions differ from RCP4.5.

RCP8.5 — **Continued rise**
High emissions keep forcing rising to about 8.5 W/m² in 2100. This is a high-end experiment, not a probability forecast.

CMIP6 / AR6 · five illustrative pathways

SSPx-y combines a societal storyline with a forcing pathway. The descriptions below refer to global net CO_2 emissions.

SSP1-1.9 — **Very low**
Rapid CO_2 cuts; net zero around 2050, followed by net-negative emissions.

SSP1-2.6 — **Low**
Strong CO_2 reductions; net zero in the second half of the century, with net-negative emissions later.

SSP2-4.5 — **Intermediate**
CO_2 remains near current levels until mid-century, then declines; no global net zero by 2100.

SSP3-7.0 — **High**
CO_2 rises, roughly doubling by 2100 amid fragmented development and weak mitigation.

SSP5-8.5 — **Very high**
Fossil-intensive growth; CO_2 roughly doubles by 2050 and remains high thereafter.

Same 2100 forcing ≠ same emissions history, peak warming or cumulative impact. Similar RCP and SSP suffixes do not make their trajectories identical.

van Vuuren et al. (2011) · RCP overview IPCC AR6 WGI · Box SPM.1 / Figure SPM.4